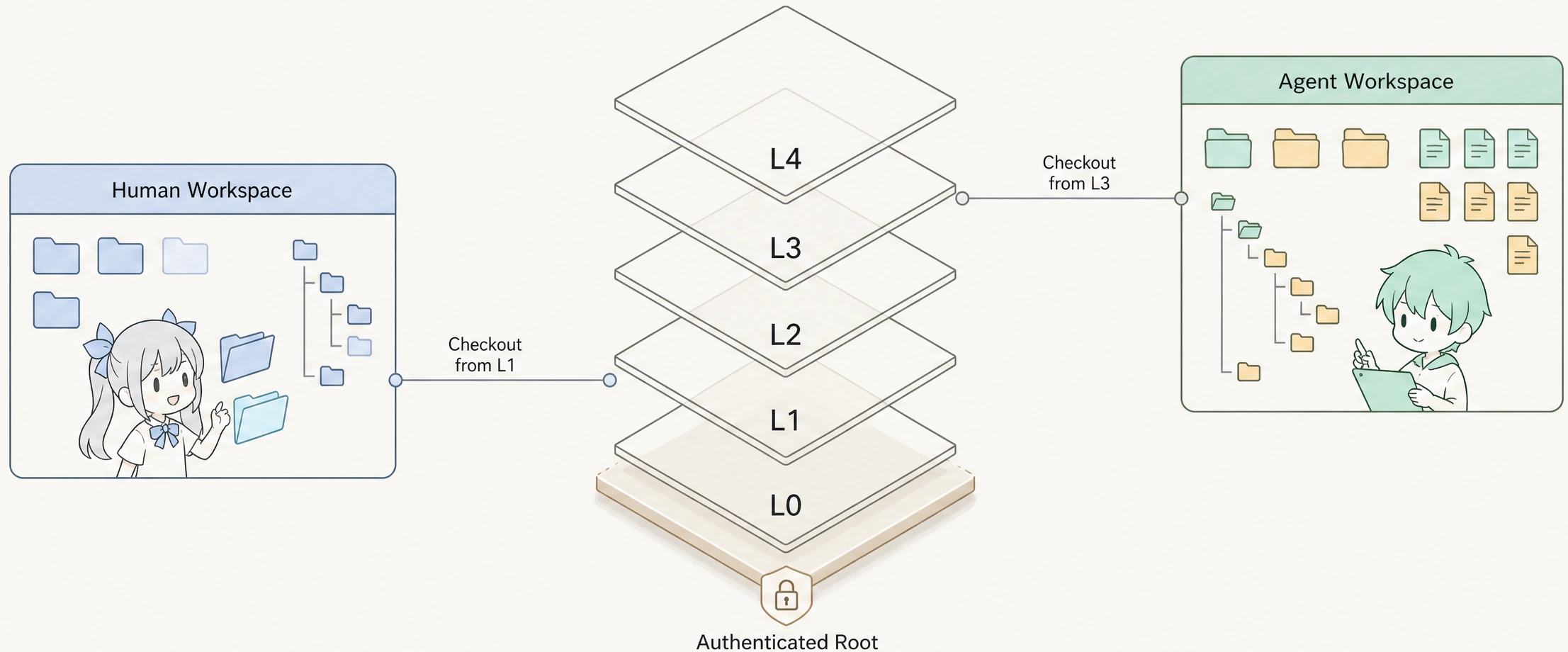


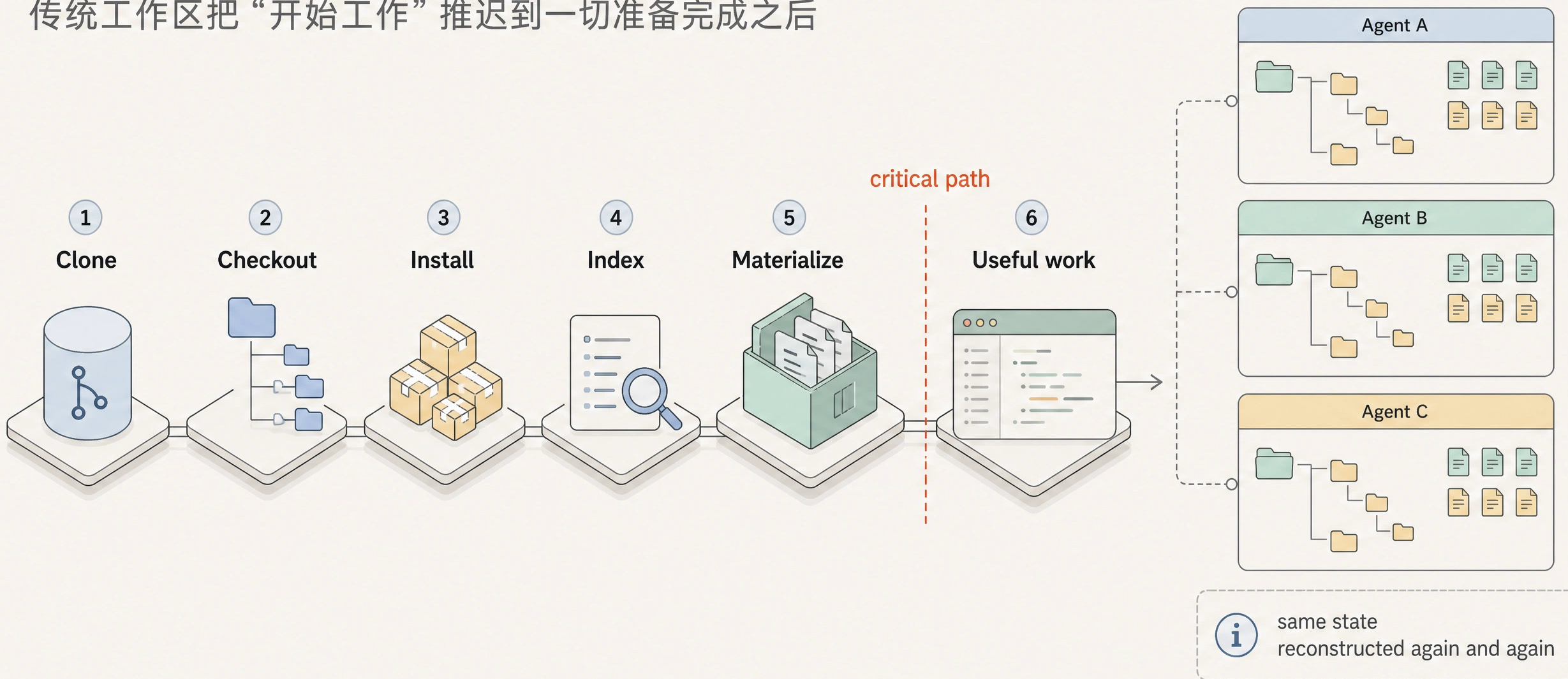
A Workspace Is a Root, Not a Directory

LayerFS：面向人类、Agents、浏览器与远程执行的通用逻辑工作区



Physical Checkout Is the Bottleneck

传统工作区把“开始工作”推迟到一切准备完成之后

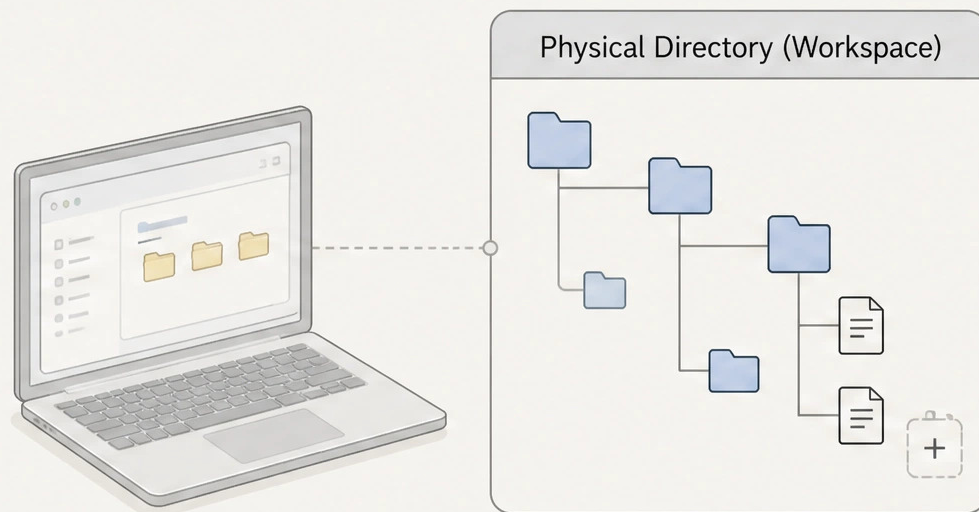


Physical to Logical

传统模型：physical directory = workspace

LayerFS：logical root = workspace

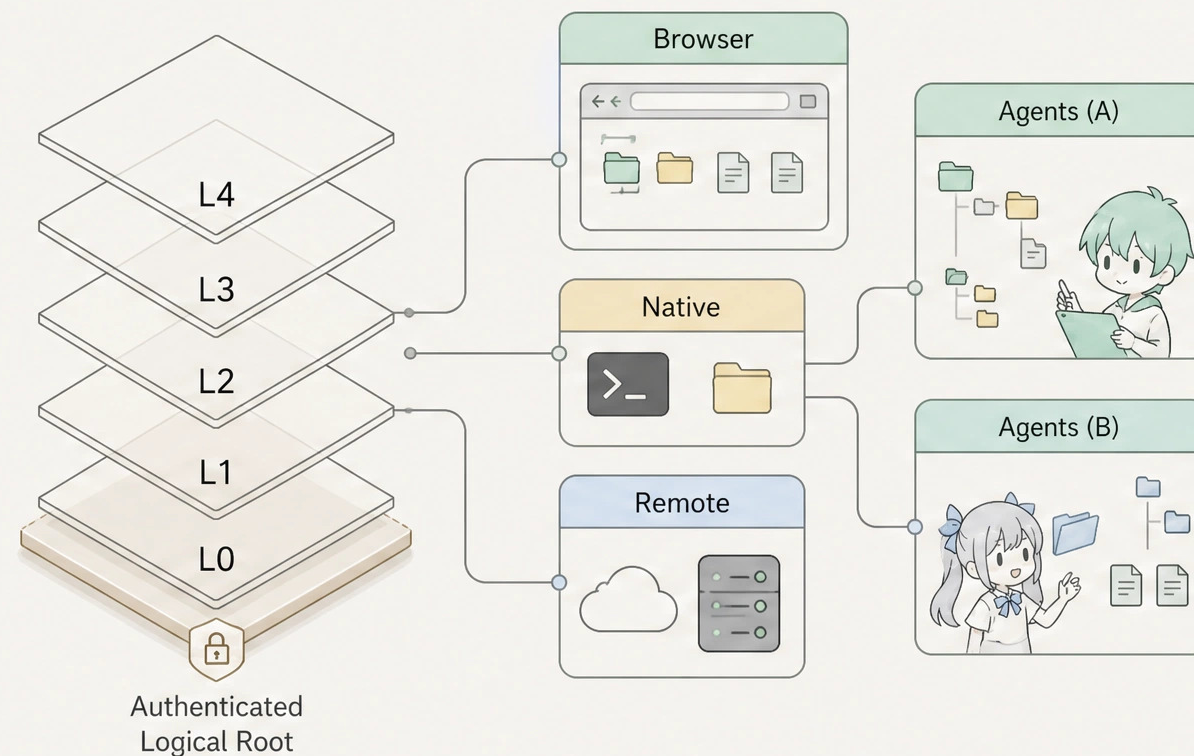
传统模型：physical directory = workspace



Local Machine

-  machine-bound
-  mutable path
-  copy to isolate

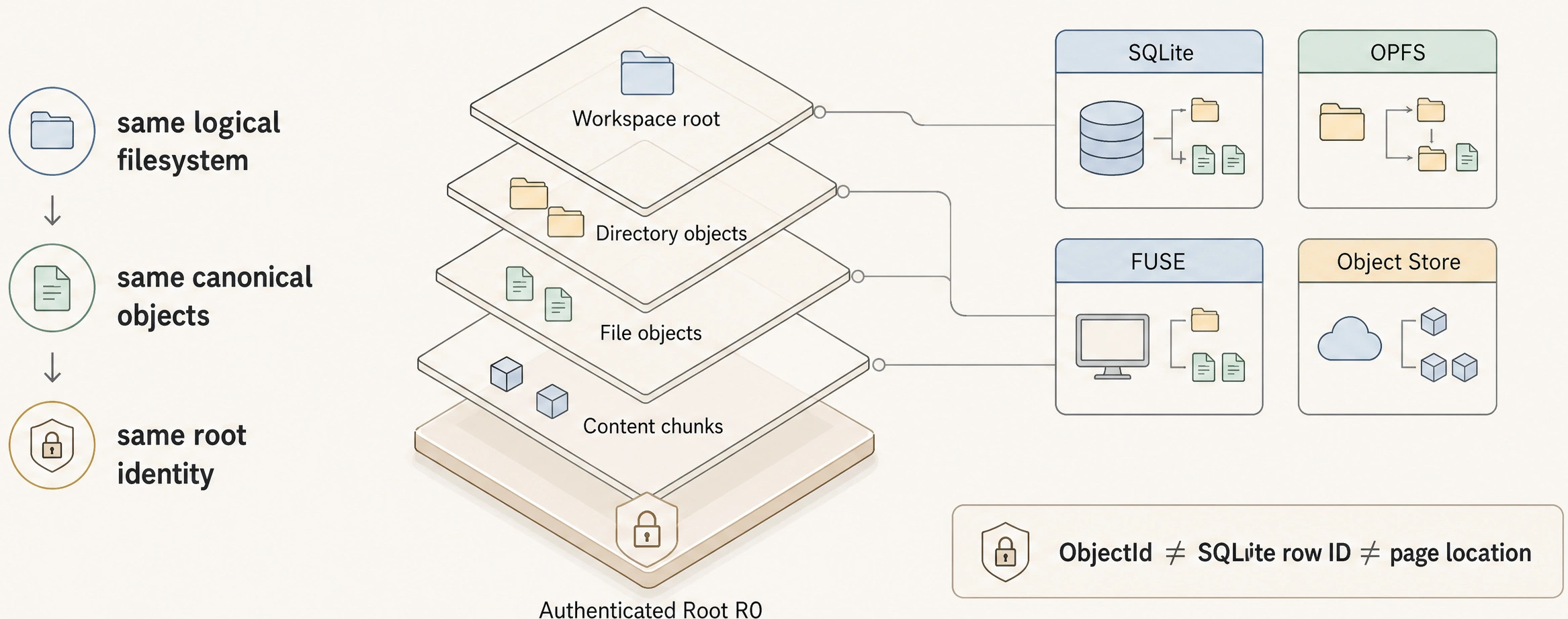
LayerFS：logical root = workspace



physical directory = optional projection

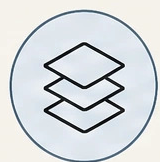
Immutable, Content-Addressed, Authenticated

The same logical filesystem keeps the same root identity across machines, databases, browsers, mounts, and storage carriers.

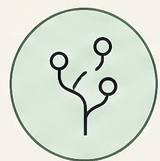


Fork Roots, Not Physical Files

Copy-on-write 让并行 Agent 变成轻量叠加层



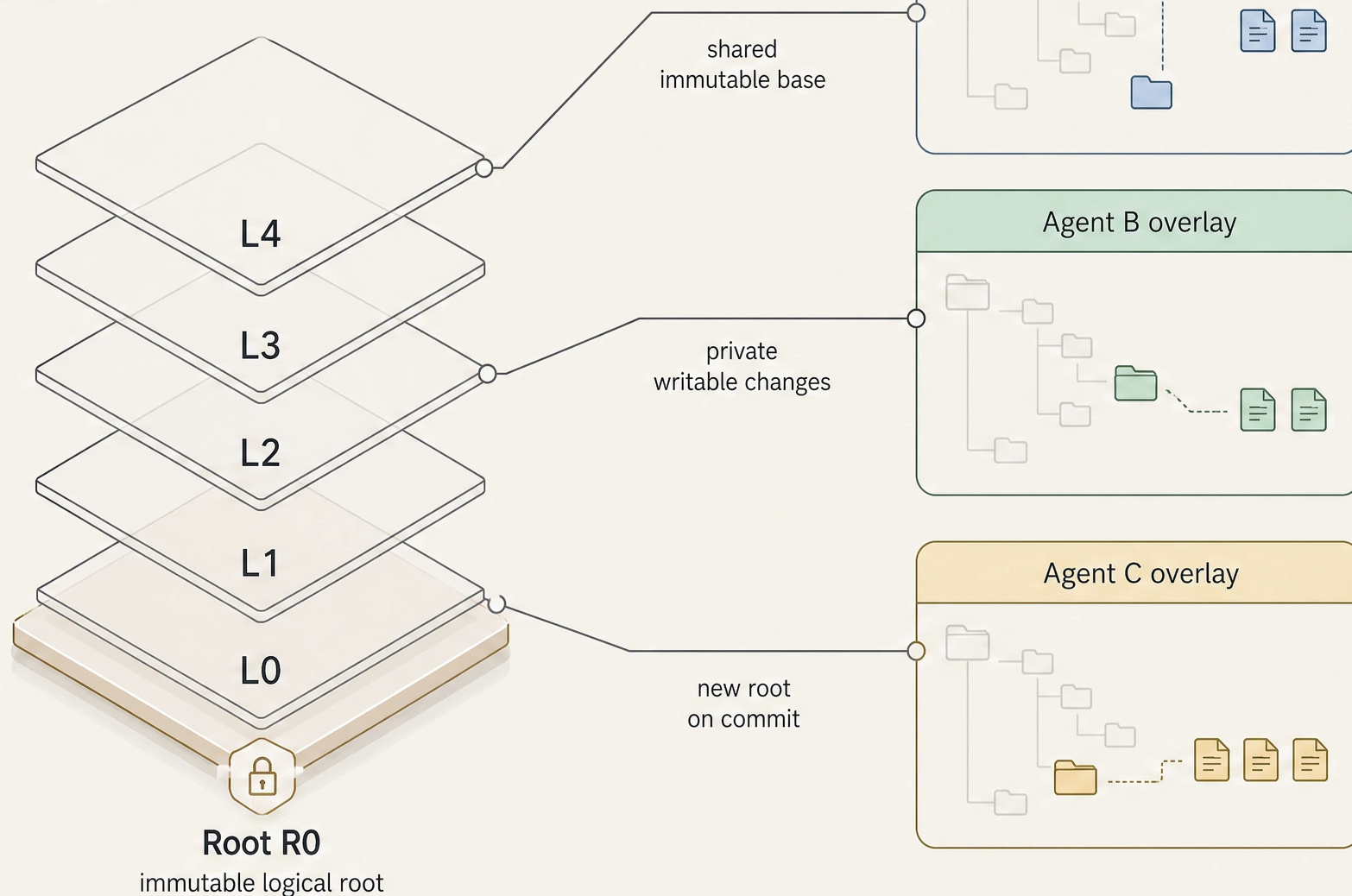
one immutable logical root
+ many lightweight
writable overlays



metadata-only
initial fork

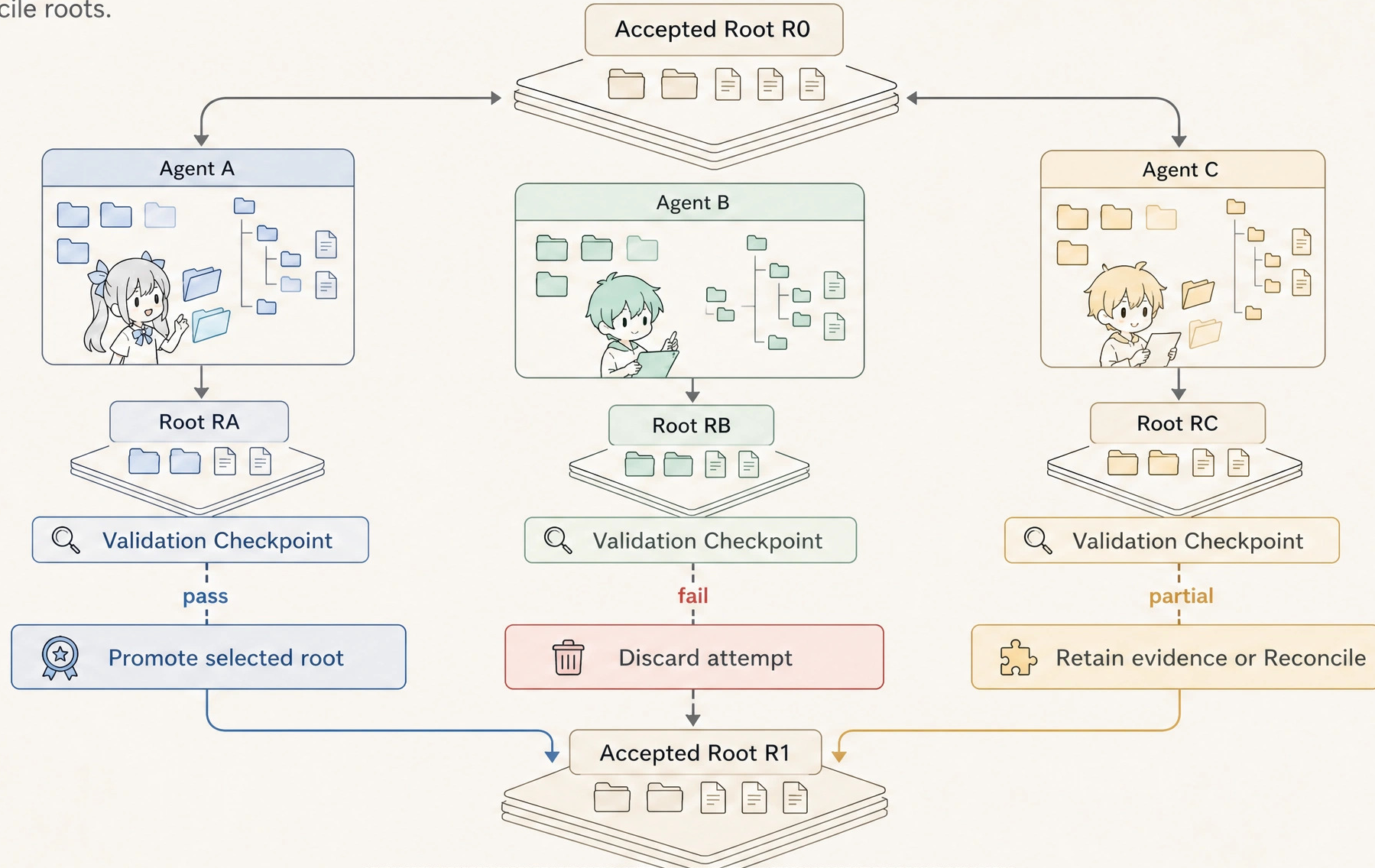


zero canonical
content copied



Coordinate by Immutable States

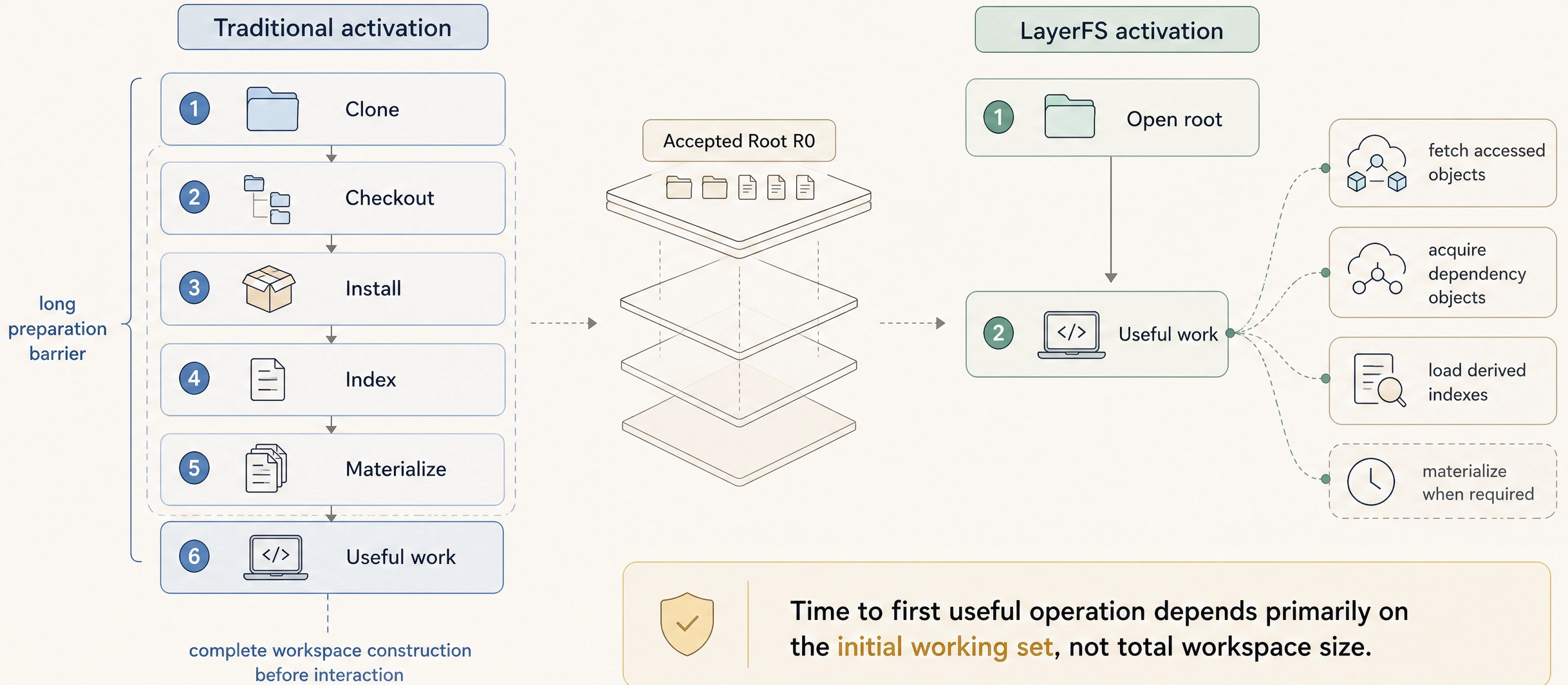
Agents do not compete for one mutable path; they compare, validate, promote, discard, and reconcile roots.



coordinate by **immutable states**, not mutable paths

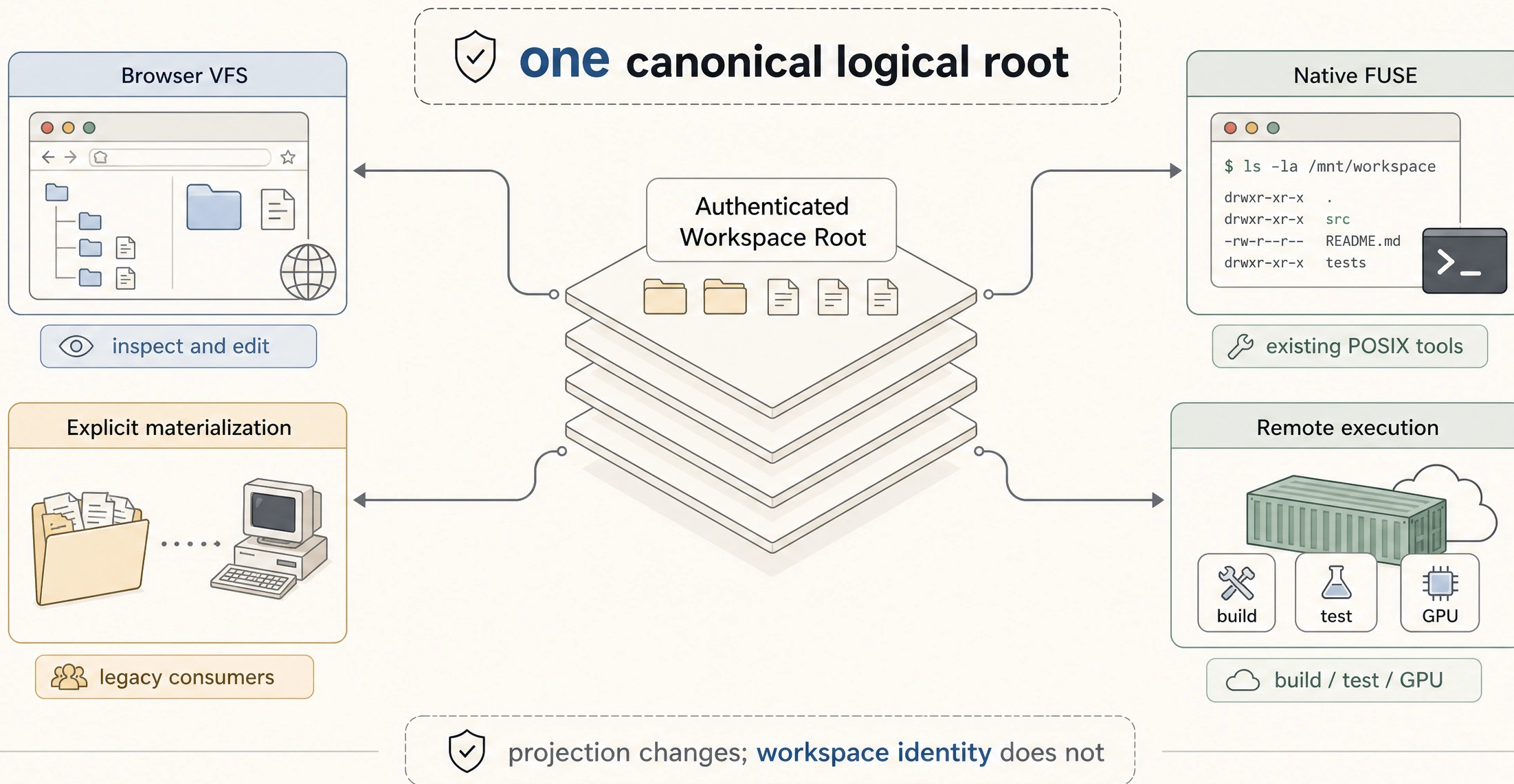
Open First, Acquire on Demand

Zero-bootstrap 不是零计算，而是移除“完整重建”的临界路径。



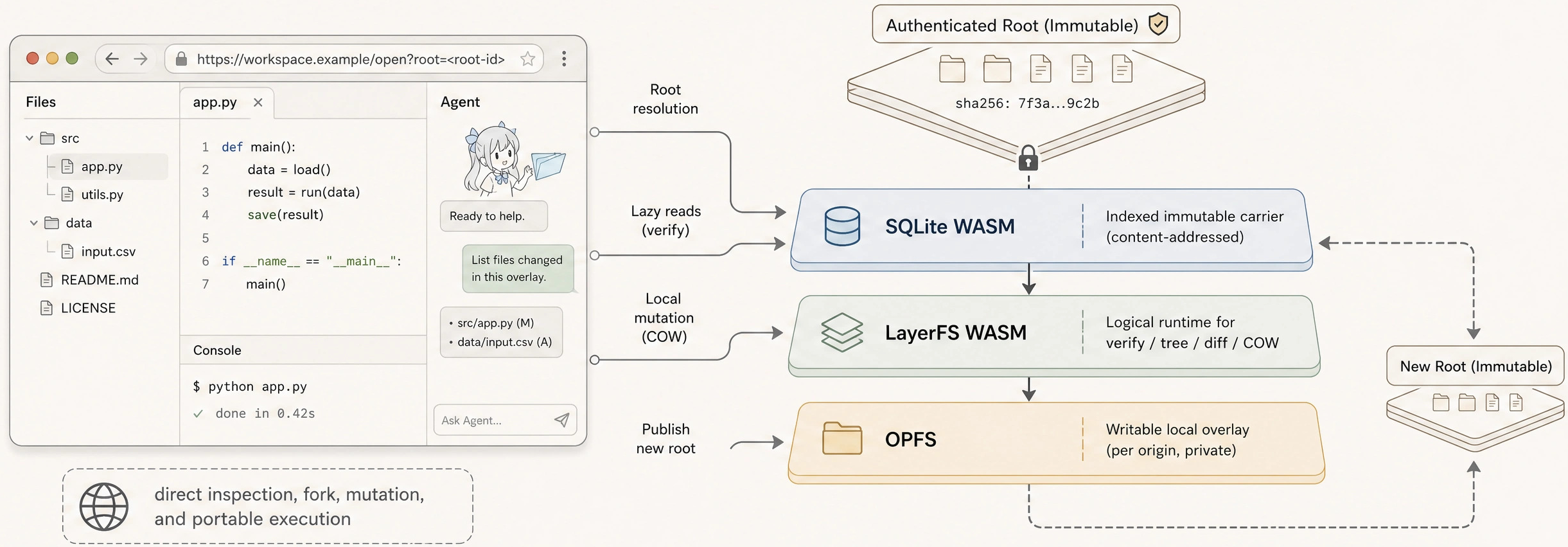
One Root, Multiple Projections

FUSE、SQLite、WASM 与远程执行不是不同的工作区



The Browser Becomes a Workspace Runtime

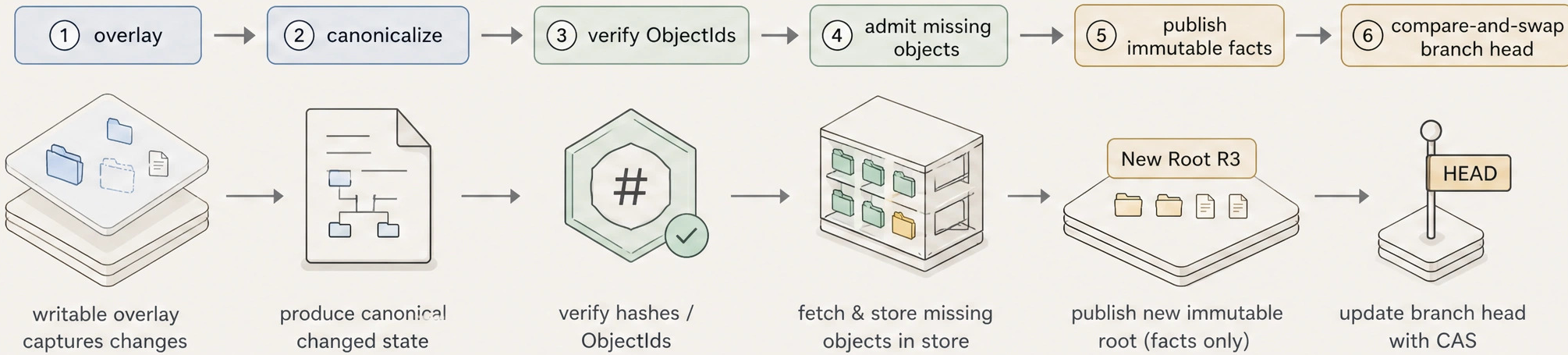
从 Repository Viewer 到一等工作区运行时



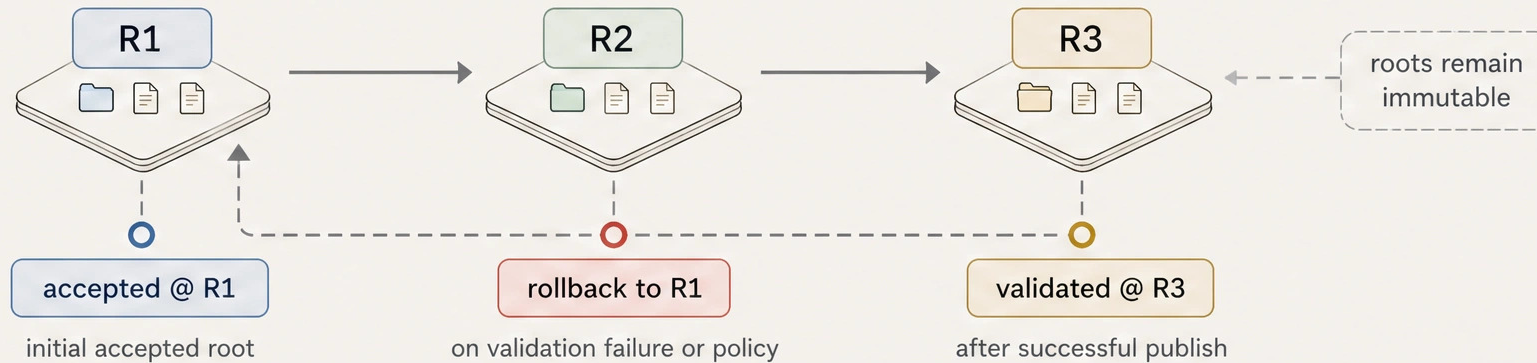
<https://workspace.example/open?root=<root-id>>

Immutable Facts, Explicit Capabilities

LayerFS publishes a new immutable root only after changed state is canonicalized, ObjectIds are verified, required objects are admitted, and the branch head is updated with compare-and-swap.



Immutable Root Chain



Capability Boundary



authenticated retrieval



bounded processing



explicit completeness



no ambient authority



folder



document



object



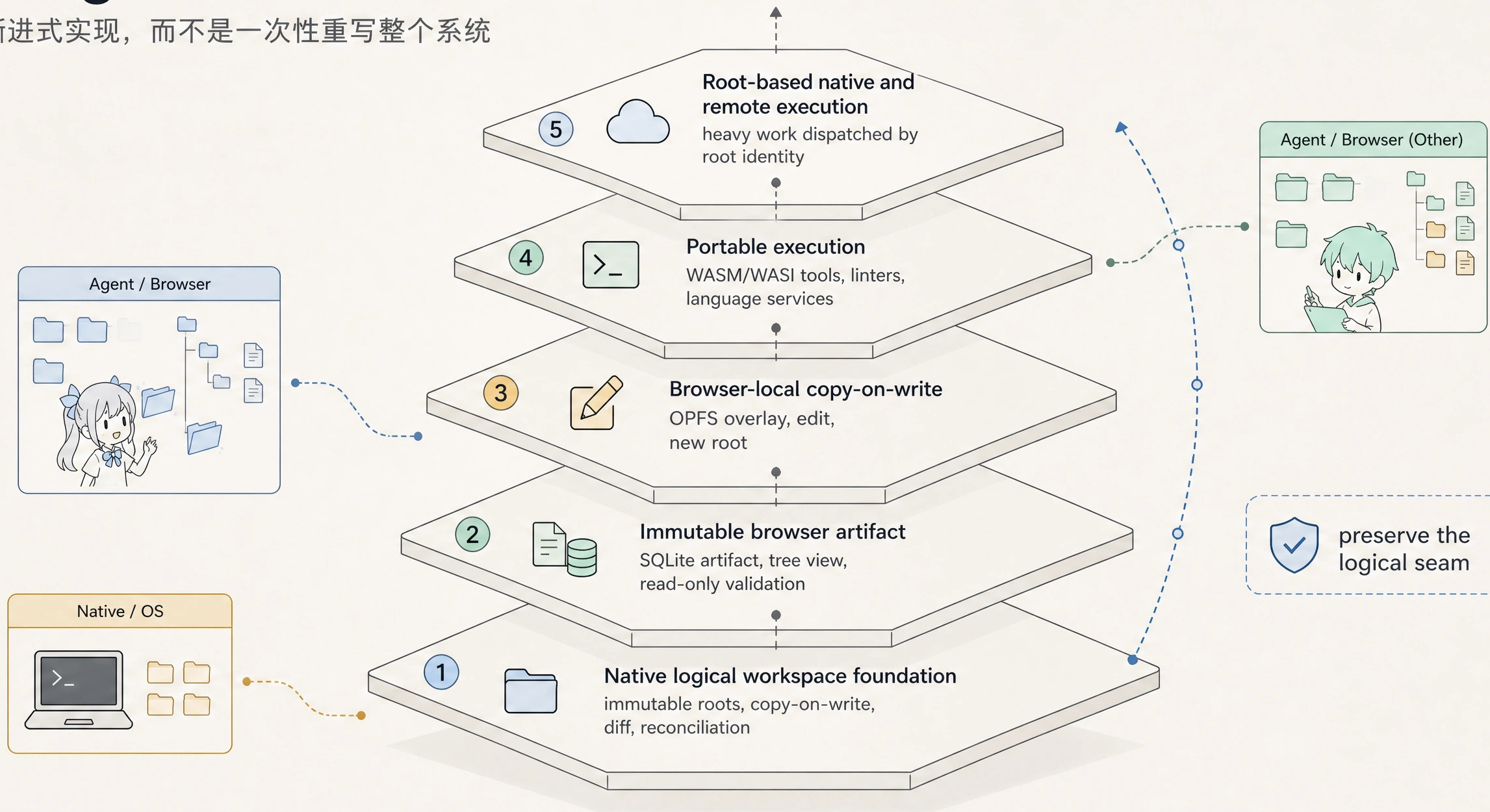
hash / ObjectId



verified

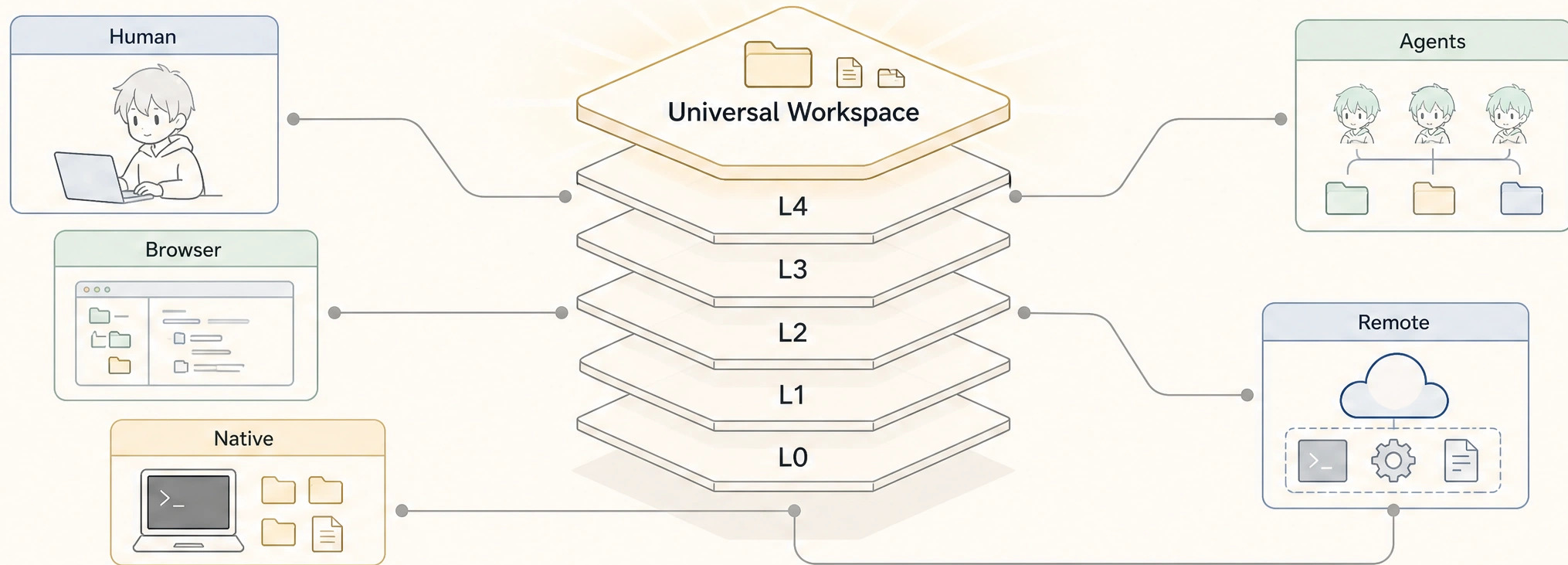
Progressive Realization

渐进式实现，而不是一次性重写整个系统



The Codebase Becomes a Universal Logical Workspace

LayerFS turns the codebase from a machine-local folder into a universal, directly usable logical workspace.



Open anywhere.



Fork safely.



Execute against the same root.